ERD CASE STUDY 2

Library System

A library system consists of a number of branch libraries. Each branch library has a name, a branch librarian, and a list of staff. Library materials can be books, periodicals, or music recordings. These materials may be duplicated either within a branch or between branches. Each branch is expected to keep track of its total number of holdings. The information for books includes a title, one or more authors, an ISBN, a publisher, and number of pages. The information for periodicals includes name, publisher, periodicity, and subscription information. For music recordings, the information includes the name of recording, the composer, the performer, the recording label, and the recording number.

Members of the library have borrowing privileges and can be general members and researchers. Researchers can borrow more materials and for a longer period than general members. Borrowers having materials overdue have to pay fine. The total number of overdue notices accumulated by a member is recorded to keep track of tardy borrowers. The number of times a particular library item is borrowed is also tracked.

The main activities of the library are catalog browsing and checking materials in and out. Users browse the on-line catalog to find out what is in library and where is it located. The catalog provides the call number of the book, the name of branch, and whether it is in the stacks at the present time. Further information about the branch, such as location and operating hours is also available. Checking-in and checking-out process are performed by staffs of the library and the checking-out date and checking-in date are kept for each transaction.